SOFTWARE ENGINEERING

PROJECT : HEALTHCARE APPOINTMENT AND PATIENT RECORD MANAGER

Centralized healthcare management for small clinics

TEAM 4

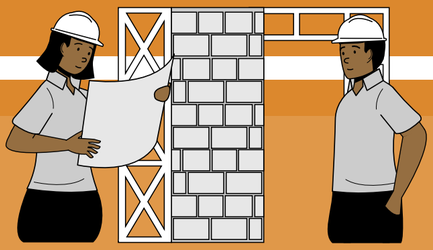
1.KARANAM SREEHITHA

2.KHUSHI PATEL

3.KIRTHANA S

4.NEHA PORLA

PROJECT OVERVIEW & PROBLEM STATEMENT



Customer / Target Users

- Small clinics and healthcare facilities.
- Administrators managing clinic operations.
- Doctors accessing patient information and medical records.
- Receptionists handling patient registration and appointments.

Problem Statement

Small clinics need an organized way to manage patient information, appointments, medical histories and prescriptions. Handling these activities through disconnected or manual processes can make record management and coordination difficult.

Our Proposed Solution

A centralized web-based application that brings essential clinic management activities together through a single system.

Project Objectives

- Simplify patient and appointment management.
- Maintain centralized medical records.
- Support prescription management.
- Provide authentication and controlled access.

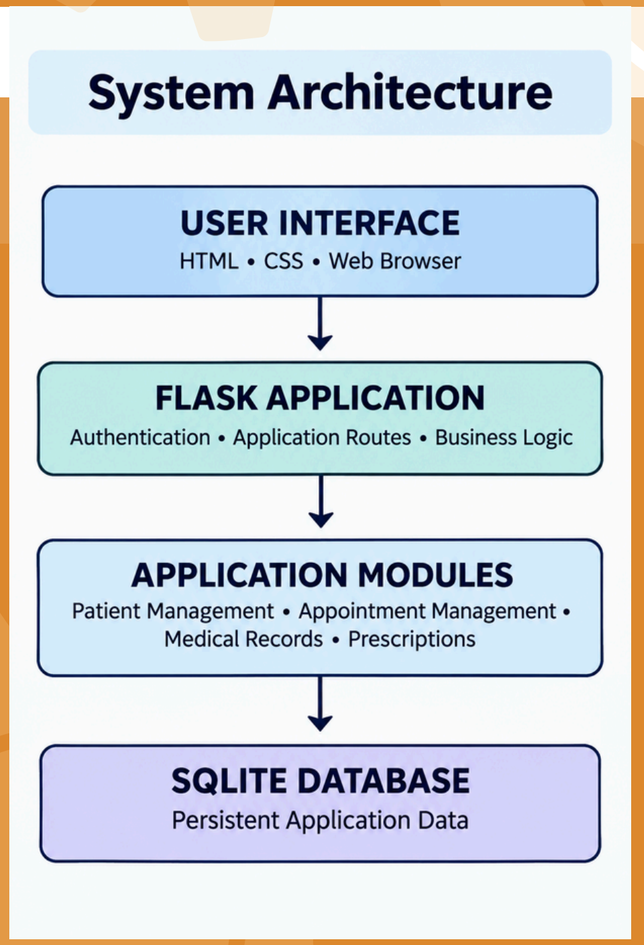
Proposed Solution & System Architecture

Module	Functionality
Patient Management	Registration, viewing, updating and searching patient records
Appointment Management	Scheduling, viewing, rescheduling and cancellation
Medical Records	Maintaining patient medical histories
Prescription Management	Managing prescriptions linked to medical records
Authentication	Login and user-role management

Technology Stack

- Frontend: HTML, CSS
- Backend: Python Flask
- Database: SQLite
- Testing: Pytest
- Version Control: Git and GitHub
- CI: GitHub Action

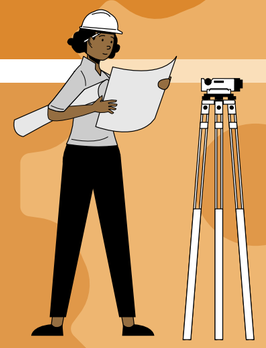






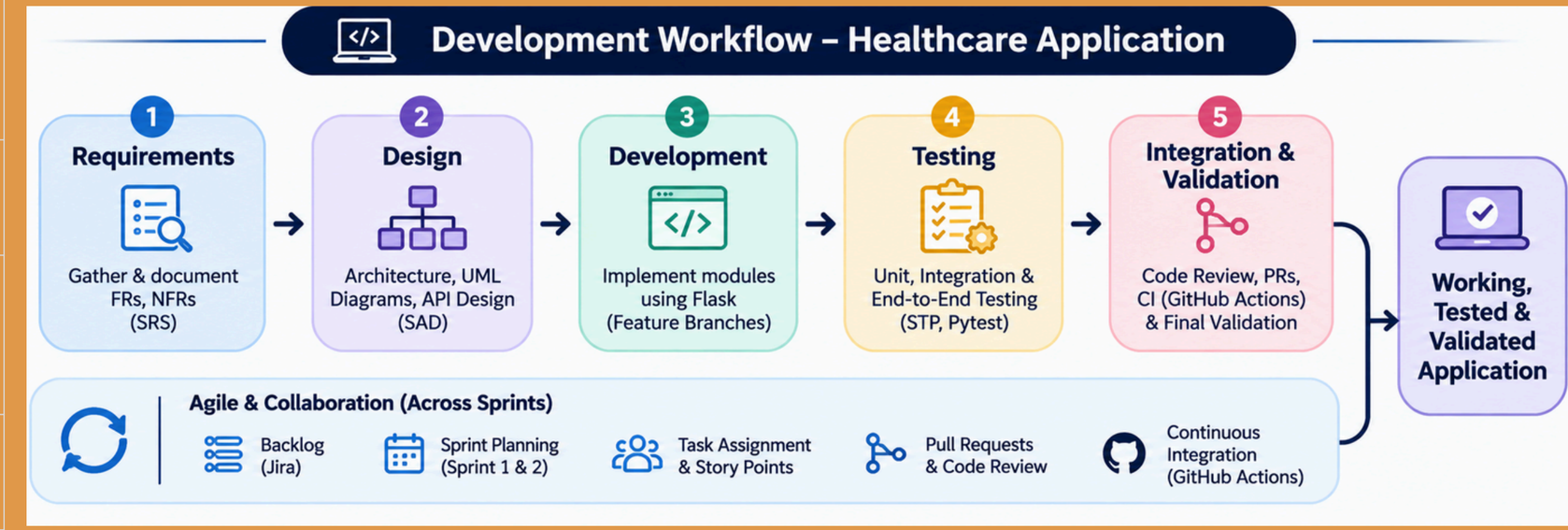
SOFTWARE ENGINEERING PRACTICES & DEVELOPMENT PROCESS

Area	Practices
Requirements	SRS and functional/non-functional requirements
Architecture	Software Architecture and Design documentation
Testing	Software Test Plan and automated Pytest testing
Agile Planning	Sprint planning, Jira tasks and story points
Collaboration	GitHub branches, pull requests and code reviews
Automation	GitHub Actions continuous integration
Documentation	SRS, SAD, STP and project README



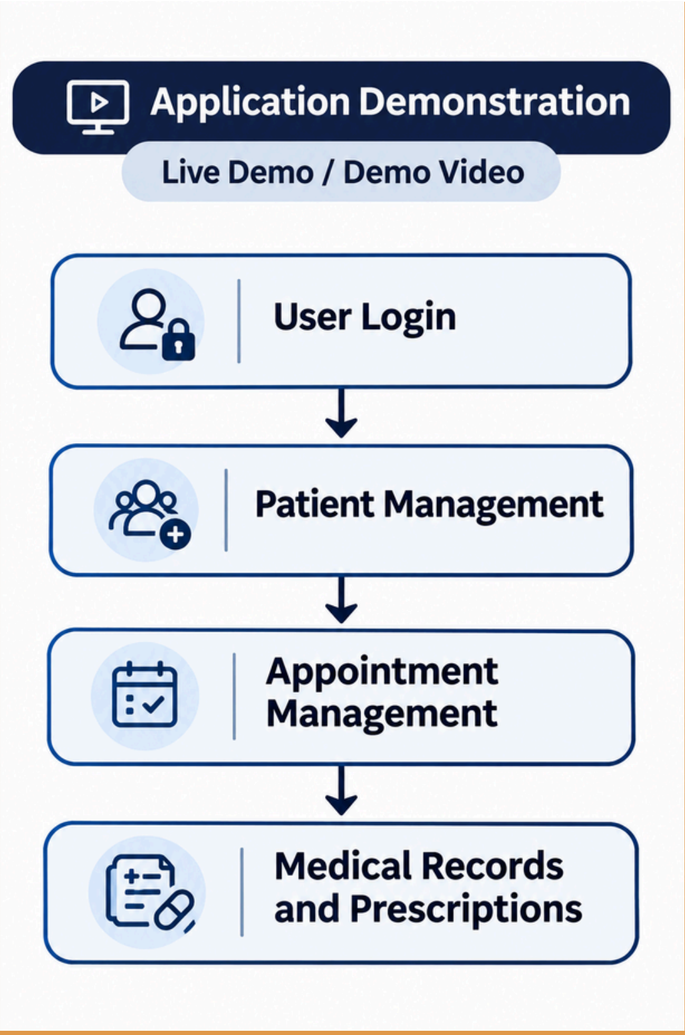
Quality and Validation

- Automated tests for application functionality.
- Authentication validation and error handling.
- Pull-request-based collaboration.
- Continuous integration through GitHub Actions.



TEAM CONTRIBUTIONS, DEMONSTRATION & CONCLUSION

Team Member	Sprint 1 – Core Development	Sprint 2 – Integration & Validation
Sreehitha	• Implemented core modules • Set up initial test cases	• End-to-end testing & validation • Final presentation & demo
Khushi	• Developed authentication & user management	• Bug fixes and application validation • Strengthened input checks
Kirthana	• Designed database schema & integrated CRUD operations	• Database and data integrity verification • Consistency and constraints validation
Neha	• Implemented remaining features & module integration	• Final integration verification • Project documentation and README

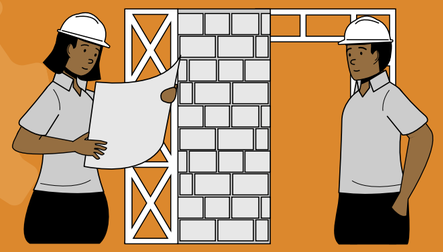


FINAL OUTCOME

- Centralized healthcare management application developed using Flask and SQLite.
- Integrated patient, appointment, medical record and prescription modules.
- Authentication and validation mechanisms implemented.
- 50 automated tests passed in the reported integrated test run.

CONCLUSION:

A MODULAR, TESTED AND INTEGRATED HEALTHCARE MANAGEMENT SOLUTION DEVELOPED THROUGH COLLABORATIVE SOFTWARE ENGINEERING PRACTICES.





THANKYOU

SREEHITHA_KHUSHI_KIRTHANA_NEHA

